सहायक प्राध्यापक (गणित) महाराष्ट्र शिक्षण सेवा, गट-अ व
अधिव्याख्याता (गणित) महाराष्ट्र शिक्षण सेवा, गट-ब
(महाविद्यालयीन शाखा)
उच्च व तंत्र शिक्षण विभाग
Assistant Professor (Mathematics) Maharashtra Education Service, Group-A
and
Lecturer (Mathematics) Maharashtra Education Service, Group-B
(Collegiate Branch)
Higher and Technical Education Department

परीक्षेचे टप्पे: **१)** लेखी परीक्षा - २०० गुण **२)** मुलाखत- ५० गुण

-: परीक्षा योजना :-

विषय व सांकेतांक	माध्यम	प्रश्नसंख्या	गुण	कालावधी	दर्जा	प्रश्नपत्रिकेचे स्वरुप
विषयांशी संबंधीत घटक (११४४)	इंग्रजी	१००	२००	एक तास	पदव्युत्तर पदवी	वस्तुनिष्ठ बहुपर्यायी

अ) नकारात्मक गुणदान -

१) प्रत्येक चुकीच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/ कमी करण्यात येतील.
२) एखाद्या प्रश्नाची एकापेक्षा अधिक उत्तरे दिली असल्यास अथवा ज्या उमेदवाराने उत्तरपत्रिकेत पूर्ण वर्तुळ चिन्हांकित केले नसेल अशा प्रश्नाचे उत्तर चुकीचे समजण्यात येऊन त्या प्रश्नाच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/कमी करण्यात येतील.
३) वरीलप्रमाणे कार्यपध्दतीचा अवलंब करताना एकूण अंतिम गुणांची बेरीज अपूर्णांकात आली तरीही ती अपूर्णांकातच राहील व पुढील कार्यवाही त्याच्या आधारे करण्यात येईल.
४) एखाद्या प्रश्नाचे उत्तर अनुत्तरित असेल तर, अशा प्रकरणी नकारात्मक गुणांची पध्दत लागू असणार नाही.

ब) अंतिम गुणवत्ता यादी ही वस्तुनिष्ठ परीक्षेतील व मुलाखतीतील एकत्रित गुणांवर आधारीत राहील.

-: अभ्यासक्रम :-

विषयांशी संबंधित घटक यामध्ये खालील घटक व उपघटकांचा समावेश असेल.

अ. क्र.	घटक व उपघटक
1	**Unit-I**
	Real Analysis: 1.1 Elementary set theory, countable and uncountable sets, real number system as a complete ordered field, Archimedean property, supremum, infimum. 1.2 Sequences and series, convergence, limsup, liminf. 1.3 Bolzano Weierstrass theorem, Heine Borel theorem. 1.4 Uniform continuity, differentiability, mean value theorem. Riemann integral 1.5 Sequences and series of functions, uniform convergence. 1.6 Functions of several variables, directional derivative, partial derivative, derivative as a linear transformation, inverse and implicit function theorem. 1.7 Lebesgue measure, Lebesgue measurable function, Lebesgue integration. 1.8 Monotonic functions, functions of bounded variations, absolutely continuous functions. 1.9 Metric spaces, compactness, connectedness, normed linear spaces, spaces of continuous functions as examples.
2	**Complex Analysis:** 2.1 Algebra of complex numbers, the complex plane 2.2 Power series, transcendental functions as exponential, trigonometric and hyperbolic functions.

	2.3 Analytic functions, Cauchy-Riemann equations. 2.4 Conformal mappings, Mobius transformations. 2.5 Contour integral, Cauchy's theorem, Cauchy's integral formula, 2.6 Liouville's theorem, Maximum modulus principle, Schwarz lemma, Open mapping theorem. 2.7 Taylor series, Laurent series, calculus of residues.
	Unit-II
3	**Algebra:** 3.1 Permutations, combinations, pigeon-hole principle, inclusion-exclusion principle, derangements. 3.2 Fundamental theorem of arithmetic, divisibility in Z, congruences, Chinese remainder theorem, Euler's Ø- function, primitive roots. 3.3 Groups, subgroups, normal subgroups, quotient groups, homomorphisms, 3.4 Cyclic groups, permutation groups, Cayley's theorem, class equations, Sylow theorem. 3.5 Rings, ideals, prime and maximal ideals, quotient rings, 3.6 Unique factorization domain, principal ideal domain, Euclidean domain. Polynomial rings and irreducibility criteria. 3.7 Fields, finite fields, field extensions, algebraic extensions, splitting fields, separable extensions, Galois groups, solvable by radicals.
4	**Topology:** 4.1 Topological spaces, basis for topology, product topology, subspace topology, closed sets and limit points, continuous functions, quotient topology. 4.2 Connectedness and compactness, countability Axioms, Separation Axioms
5	**Probability** 5.1 Introduction to probability, probability space, events, sigma fields, sigma field of borel sets, limit superior and limit inferior for a sequence of events. Probability measure, conditional probability, Bayes formula. 5.2 Random variables, discrete and absolutely continuous random variables, Bernoulli, Binomial poisson and normal distributions, expectation and variance of random variables both discrete and absolutely continuous. 5.3 Characteristic functions, Chebyshev inequality, weak law of large numbers, Convergence of random variables, Kolmogorov strong law of large numbers, Central limit theorem
	Unit-III
6	**Ordinary differential Equation(ODE)** 6.1 Existence and uniqueness of solutions to initial value problem of first order ordinary differential equations (ODE), system of first order linear ODE, with constant coefficient and variable coefficients. 6.2 Linear dependence and independence of solution of homogenous n-th order linear ODE, Wronskian matrix, Lagrange's method (variation of parameters). 6.3 Sturm-Liouville separation and comparison theorems, oscillation properties of solutions, Eigen values and eigen functions of sturm-Liouville Boundary value problem.
7	**Partial differential Equation(PDE)** 7.1 Lagrange and Charpit methods for solving first order Partial Differential Equations (PDE), Cauchy's problem for first order PDEs. 7.2 Classification of second order PDEs, General Solution of higher order PDEs with constant coefficients, Method of separation of variables for Laplace, Heat and Wave equations.
8	**Calculus of Variations** 8.1 The variation of a functional, Euler-Lagrange equation, necessary and sufficient conditions for extrema, The case of several variables, simple variable end point problem. 8.2 Variational problems with subsidiary conditions, variational methods for boundary value problems in ordinary and partial differential equations.
9	**Classical Mechanics** 9.1 Generalized coordinates, Lagrange's equations, Hamilton's canonical equations, Hamilton's principle and principle of least action. 9.2 Two dimensional motion of rigid bodies, Euler's dynamical equations for the motion of a rigid body about an axis, theory of small oscillations.

	Unit IV
10	**Linear Algebra** 10.1 Vector Spaces over a field F, Subspaces, Basis and dimensions. Linear Transformations, rank and nullity of a linear transformation. 10.2 Matrix associated with a linear transformation, rank of a matrix and its equivalence to rank of linear transformation. Determinant of matrices,linear equations. 10.3 Characteristic polynomial, annihilating polynomial, Cayley Hamilton Theorem, minimal polynomial, eigen values and eigen vectors. 10.4 Invariant subspaces, diagonalization, triangularization Jordan Canonical form. 10.5 Inner product spaces, orthonormal basis. 10.6 Quadratic forms, reduction and classification of quadratic forms.
11	**Linear Integral Equations** 11.1 Linear integral equation of the first and second kind of Fredholm and Volterra type, Solutions with separable kernels. Characteristic numbers and eigen functions, resolvent kernel. 11.2 Types of Integral Equations: Initial value problems, Boundary value problems, Singular integral equations, Abel integral equations.
12	**Numerical Analysis** 12.1 Numerical solutions of algebraic equations, method of iterations and Newton-Raphson method, rate of convergence. 12.2 Solution of a system of linear algebraic equations using Gauss elimination and Gauss-Seidel methods. 12.3 Finite differences, Lagrange, Hermite and spline interpolation. 12.4 Numerical Differentiation and Integration. 12.5 Numerical solutions of ODEs using Picard, Euler, modified Euler and Runge-Kutta methods.

दिनांक – ३० एप्रिल, २०२५

अवर सचिव
महाराष्ट्र लोकसेवा आयोग